

**PH 101**  
**Tutorial 2**  
**Date: 17/08/2023**

---

**Q.1-Q.7 are to be discussed in the Tutorial class.**

1. A particle moves with constant velocity  $v$  along the curve  $r = k(1 + \cos \theta)$ . Find and write in terms of  $v, k, \theta$  (as applicable):

(a)  $\vec{a} \cdot \hat{r}$

(b)  $\vec{a} \cdot \hat{\theta}$

(c)  $|\vec{a}|$

Here  $\vec{a}$  refers to acceleration.

2. A bird flies on a spiral trajectory such that its polar coordinates at time  $t$  are given by:

$$r = Ae^{\omega t}, \theta = \omega t$$

Here  $A$  and  $\omega$  are positive constants. Find the velocity ( $\vec{v}$ ) and acceleration ( $\vec{a}$ ) vectors of the bird at time  $t$ . Show that the angle between these vectors, i.e.  $\vec{v}$  and  $\vec{a}$  is always  $\frac{\pi}{4}$ .

3. A particle moves outward along a spiral. Its trajectory is given by  $r = A\theta$ , where  $A$  is a constant.  $A = (1/\pi)$  m/rad.  $\theta$  increases in time according to  $\theta = \alpha t^2/2$ , where  $\alpha$  is a constant.

(a) Sketch the motion, and indicate the approximate velocity and acceleration at a few points.

(b) Show that the radial acceleration is zero when  $\theta = 1/\sqrt{2}$  rad.

(c) At what angles do the radial and tangential accelerations have equal magnitude?

4. Consider two concentric cylinders, centred on the vertical  $z$ -axis, with radii  $R \pm a$ , where  $a$  is very small. A small frictionless hard rubber disc of thickness  $2a$  is inserted between the two cylinders, so that it can be considered a point mass that can move freely at a fixed distance from the vertical axis. Using cylindrical coordinate system, write down and solve Newton's second law for the general motion of the rubber disc, including the effects of gravity. Describe the motion of the rubber disc.

5. An insect flies on a trajectory such that its polar coordinates at time  $t$  are given by

$$r = \frac{bt}{\tau^2}(2\tau - t), \theta = t/\tau, (0 \leq t \leq 2\tau). \text{ Here } b \text{ and } \tau \text{ are positive constants.}$$

Find the velocity vector of the insect at time  $t$ . Show that the least speed achieved by the insect is  $b/\tau$ . Find the acceleration of the insect at this instant.

6. Show that the angular momentum  $L$  and the energy  $E$  of a particle of mass  $m$  moving in a potential  $V(r)$  can be expressed in terms of plane polar coordinates as

(a)  $\vec{L} = mr^2\dot{\theta}\hat{z}$ ,

(b)  $E = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + V(r)$

Here,  $\hat{z} = \hat{r} \times \hat{\theta}$  is a unit vector perpendicular to the plane of the trajectory.

7. Convert the coordinates as indicated

(a)  $(8, \frac{\pi}{4}, \frac{\pi}{6})$  from spherical to Cartesian.

(b)  $(2\sqrt{3}, 6, -4)$  from Cartesian to cylindrical.

### Practice Problems

8. (a) Find the unit vectors  $\hat{r}$ ,  $\hat{\theta}$  and  $\hat{\phi}$  of a spherical coordinate system in terms of  $\hat{x}$ ,  $\hat{y}$  and  $\hat{z}$ .

(b) Solve  $\hat{x}$ ,  $\hat{y}$  and  $\hat{z}$  in terms of  $\hat{r}$ ,  $\hat{\theta}$  and  $\hat{\phi}$ .

9. Prove that a spherical coordinate system is orthogonal.

10. Convert the coordinates as indicated

(a)  $(3, \frac{\pi}{3}, -4)$  from cylindrical to Cartesian.

(b)  $(-2, 2, 3)$  from Cartesian to cylindrical.